

UAC Predictive Intelligence System: Forecasting Child Welfare Demand Using Time-Series Analytics and Machine Learning

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Abstract

This project develops a predictive intelligence framework for forecasting children in HHS care, predicting discharge demand, and identifying capacity stress using time-series forecasting and machine learning techniques.

Introduction

The U.S. Department of Health and Human Services manages the Unaccompanied Alien Children Program. Demand fluctuates due to migration trends, policy changes, and humanitarian crises. Predictive analytics can support proactive planning.

Objectives

- Forecast children in HHS care
- Predict discharge demand
- Identify capacity stress
- Provide early warning indicators

Methodology

Data preprocessing, feature engineering, time-series forecasting, ARIMA/SARIMA models, Random Forest, Gradient Boosting, and performance evaluation using MAE, RMSE, and MAPE.

Results and Discussion

Analysis indicates that predictive modeling can improve planning accuracy and help allocate resources proactively.

Conclusion

The proposed framework transforms historical reporting into predictive intelligence and supports better child-welfare outcomes.

References

1. Forecasting: Principles and Practice
2. Time Series Analysis: Forecasting and Control
3. Random Forests (Breiman, 2001)
4. Gradient Boosting Machine (Friedman, 2001)